

Minnesota Code Addresses Depressurization

Builders in Minnesota say they generally agree with the need for mechanical ventilation, but they are not so understanding about problems caused by depressurization (see "Putting Pressure on Building Codes," *HE* Sept/Oct '96, p. 39). Although several thousand new homes in Minnesota have incorporated mechanical ventilation over the past 20 years, depressurization has not received serious consideration.

Now Minnesota has adopted a new energy code that mandates tight construction and specifically addresses mechanical ventilation, as well as protection against excessive depressurization and the problems it can cause—such as backdrafting or radon infiltration. The code requirements regarding these issues take effect April 15, 2000, and are limited to detached one- and two-family dwellings. Min-

nesota's new code builds on ventilation standards currently existing in Washington and Oregon, but it is the first state code to include depressurization standards (although there are similar, less stringent regulations in the 1995 Building Code of Canada).

Addressing depressurization is especially important because atmospherically vented combustion appliances are still allowed under the new code. Sealed-combustion appliances are widely accepted as the solution to backdrafting problems, but the need for makeup air when large spot exhaust fans are installed seems harder to accept. The impact of more powerful and increasingly popular exhaust appliances on house depressurization—which is overlooked by the mechanical codes—is now finally addressed in Minnesota's building code.

all others. Although heat recovery equipment is not required, if it is installed, it must be rated and labeled.

Ventilation systems must be designed and installed to permit the required air flow. Here the code contains a performance standard—above a certain minimum air flow (30 CFM), ventilation system air flow into and out of the house at air intakes or exhaust points must be validated by measurement.

Six Paths for Depressurization

For protection against excessive depressurization, the code requires a builder to follow one of six optional compliance methods: four prescriptive paths, one aggregate path based on the four prescriptive paths, and one performance path.

The qualifying criterion for each of the four prescriptive paths (Paths 0–3) is the depressurization tolerance of the least tolerant vented-combustion appliance. (See Figure 1.) Path 0 applies where all vented-combustion appliances are sealed combustion. Path 1 applies where there is a vented-combustion appliance that is power vented or direct vented. Path 2 applies where there is an atmospherically vented furnace. Path 3 applies where there is an atmospherically vented water heater. The four prescriptive paths designate makeup air requirements for each of the categories of mechanical exhausting devices: the kitchen exhaust, the clothes dryer, and other exhausts. The quantity of makeup air available by infiltration and passive opening is limited for each of the prescriptive paths; makeup air needed in excess of these amounts must be made up by a powered makeup air source (see Figure 2).

The aggregate prescriptive path includes a table that gives the sum of the makeup air required for each of the four prescriptive paths without listing specific associated mechanical exhaust appliances.

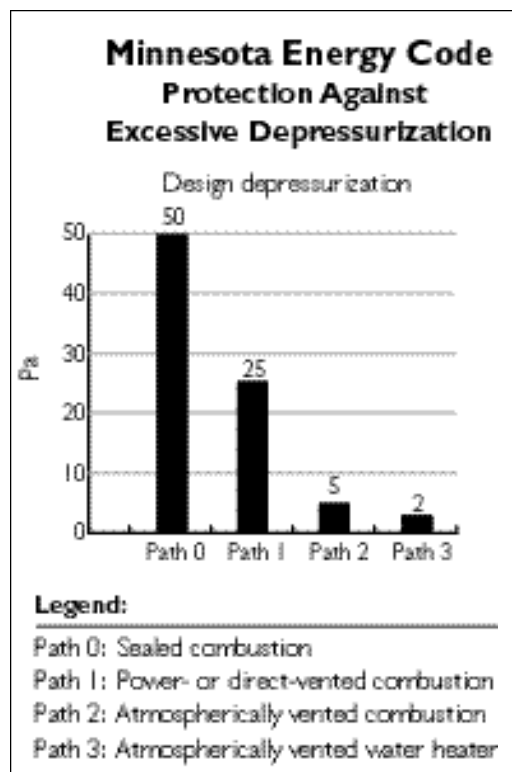


Figure 1. The path designation for the home is determined by the depressurization tolerance of the installed vented combustion equipment.

Ventilation Provisions

The mechanical ventilation provisions are in three parts: ventilation air quantity, equipment, and system design and installation. The requirement for fan-powered supply of outdoor air is 0.35 ACH, which is adopted directly from ASHRAE Standard 62-1989. Because this quantity may be exceeded if the fan is run continuously, the code also follows ASHRAE in defining a "people" ventilation rate (called "principal" ventilation in other standards) of 15 CFM per bedroom plus an additional 15 CFM. Minnesota specifies a minimum of 45 CFM, total.

This rate is intended for continuous operation whenever the house is occupied. The equipment providing this rate must meet the following specifications: Fans must be rated for continuous operation and must be quiet—1 sone for surface-mounted fans, and 1.5 sones for

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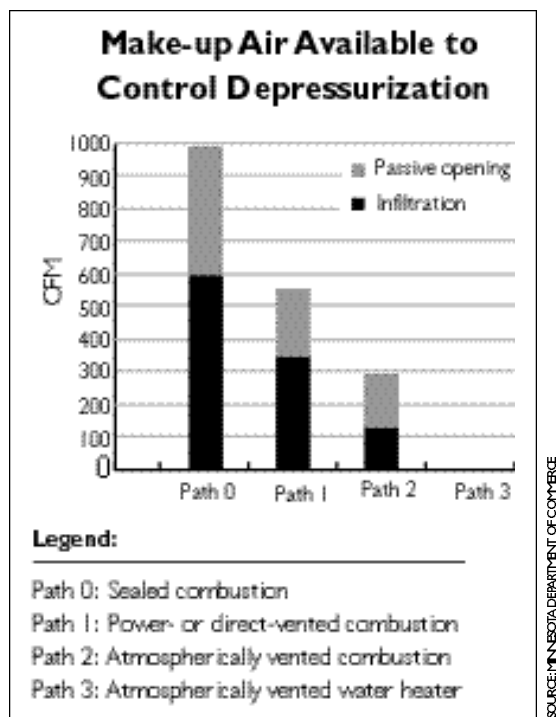


Figure 2. The quantity of makeup air available by infiltration and passive openings is limited by the depressurization tolerance (indicated by the path) in the home.

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The performance path is based on the Canadian General Standards Board Standard 51.71, "The Spillage Test." The code includes prescriptive compliance paths to make depressurization provisions acceptable to builders and inspectors.

Most readers of *Home Energy* would probably be quick to point out the superiority of a performance approach to this issue, with its simplicity and its ability to account for complex variables. Those who execute and enforce the code, however, overwhelmingly favor prescriptive measures, so we put them in the new code to make sure that the provisions are effectively implemented. The prescriptive paths do not weaken the requirements for avoiding depressurization.

—Bruce D. Nelson

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For more information:

The Minnesota Energy Code is available in electronic format at two locations on the Internet. Adobe Acrobat versions can be downloaded from the Minnesota Department of Commerce site at www.dpsv.state.mn.us (look for the Energy Code page), and an HTML version is available at www.revisor.leg.state.mn.us/arule/7672.